

Integrating PfSense with Wazuh SIEM

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Introduction

A **firewall** is a critical security solution that inspects and controls incoming and outgoing traffic on a network or device. Think of it like a **security guard at the entrance of a building**: it decides who can enter and who must stay out, based on predefined rules.

In modern environments, firewalls help:

- Prevent unauthorized access
- Block malicious traffic
- Segment networks for better security

There are different types of firewalls: **packet-filtering, stateful inspection, proxy-based, and next-generation firewalls (NGFWs)**, each providing increasing levels of inspection and control.

In this guide, we build a **hands-on SOC lab** using:

- **pfSense** → an open-source firewall and router.
- **Wazuh** → an open-source SIEM for log analysis, alerting, and threat detection.

By integrating pfSense logs into Wazuh, we simulate **real-world SOC operations**, monitoring both network and endpoint events.

1 Step 1: Requirements

Before starting, ensure you have the following:

1. **Virtual Machine Software** → VMware Workstation Pro (used in this guide, but VirtualBox also works).
2. **pfSense ISO** → Download the latest version from the official pfSense website.
3. **Wazuh Manager** → Running and accessible in your lab environment.

2 Step 2: Creating a Virtual Machine for pfSense

2.1 Open VMware

Navigate to **File** > **New Virtual Machine**.

2.2 Select the Installation Media

Choose **Installer disc image file (ISO)** and browse for the downloaded pfSense ISO.

2.3 Name Your VM

Use a descriptive name such as **pfSense** and pick a save location.

2.4 Allocate Disk Space

- **20 GB minimum** (store as a single file recommended).

2.5 Customize Hardware • RAM: 1 GB

minimum (2 GB recommended).

- **CPU:** 1 vCPU (2 cores recommended).
- **Network Adapters:**
 - **Adapter 1 (WAN)** → NAT (simulates internet access).
 - **Adapter 2 (LAN)** → Host-only (isolated internal lab network).

2.6 Assign Interfaces

- **WAN (em0)** → Configured via DHCP (IP in 192.x.x.x/24 range).
- **LAN (em1)** → Static IP 192.168.1.1/24.

```
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: [REDACTED]/24
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces         10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system             14) Enable Secure Shell (sshd)
6) Halt system               15) Restore recent configuration
7) Ping host                 16) Restart PHP-FPM
8) Shell

Enter an option: 
```

Figure 1: pfSense interface

3 Step 3: Accessing the pfSense Web Interface

1. Open a browser from a VM (e.g., Kali Linux) connected to the LAN network.
2. Go to: <https://192.168.1.1/>
3. Log in with default credentials:
 - Username: admin
 - Password: pfsense

Important: Change the default credentials immediately after login.

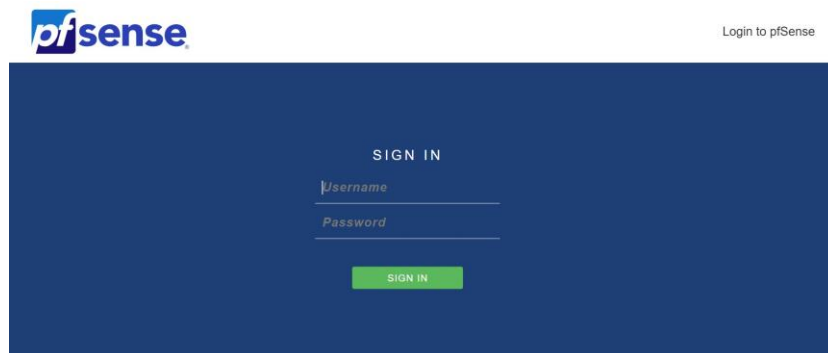


Figure 2: pfSense web interface login

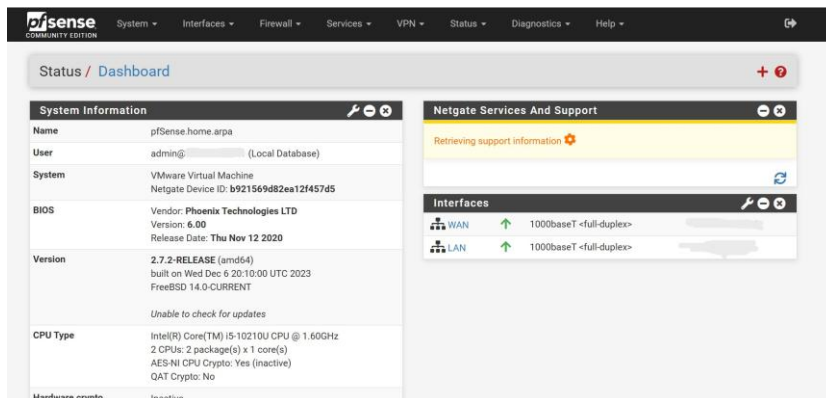


Figure 3: pfSense dashboard

4 Step 4: Why Combine pfSense with Wazuh?

- **pfSense** acts as the **network security gateway**, generating logs on traffic, connections, and blocked events.
- **Wazuh** functions as the **Security Information and Event Management (SIEM)** platform, collecting and analyzing logs to detect suspicious activity.

Together, they form a **mini SOC environment**, where firewall events are monitored and correlated with endpoint activity.

5 Step 5: Configuring pfSense Logs in Wazuh

5.1 Enable Remote Logging in pfSense

- Go to **Status** → **System Logs** → **Settings**.
- Check **Enable Remote Logging**.
- Add your **Wazuh server's IP** (e.g., 192.168.200.50) under **Remote Syslog Servers**.
- Set **Port** to 514 (default syslog).
- Select log categories to forward: System, Firewall, VPN, DHCP, DNS.

Remote Logging Options

Enable Remote Logging ☒ Send log messages to remote syslog server

Source Address Default (any)
This option will allow the logging daemon to bind to a single IP address, rather than all IP addresses. If a single IP is picked, remote syslog servers must all be of that IP type. To mix IPv4 and IPv6 remote syslog servers, bind to all interfaces.
 NOTE: If an IP address cannot be located on the chosen interface, the daemon will bind to all addresses.

IP Protocol IPv4
This option is only used when a non-default address is chosen as the source above. This option only expresses a preference; If an IP address of the selected type is not found on the chosen interface, the other type will be tried.

Remote log servers 514 IP[port] IP[port]

Remote Syslog Contents ☒ Everything
☐ System Events
☐ Firewall Events
☐ DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns)
☐ DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client)
☐ PPP Events (PPPoE WAN Client, LZTP WAN Client, PPTP WAN Client)

Figure 4: pfSense remote logging configuration

5.2 Configure Wazuh Syslog Input

On the Wazuh Manager, edit `/var/ossec/etc/ossec.cfg`:

```
<ossec_config>
  <!-- pfSense syslog input -->
  <remote>
    <connection>syslog</connection>
    <port>514</port>
    <protocol>udp</protocol>
    <allowed-ips>YOUR_PF_SENSE_IP</allowed-ips>
    <local_ip>YOUR_WAZUH_SERVER_IP</local_ip> </remote>
</ossec_config>
```

Restart Wazuh after changes.

6 Step 6: Creating a Custom Decoder and Rules for pfSense Logs

Since pfSense logs have a unique format, Wazuh may not parse them correctly out of the box.

6.1 Custom Decoder

A **decoder** extracts key fields (source IP, destination IP, protocol).

Example (pfsense-custom-decoder.xml):

```
<decoder name="pfsense-custom">  
    <prematch>filterlog</prematch>  
</decoder>  
  
<decoder name="pfsense-fields">  
    <parent>pfsense-custom</parent>  
        <regex>(\\w+)|d+:\\S*,\\S*,\\S*,(\\S+),\\S*,\\S*,(\\S+),\\S*,\\S*,\\S  
            *,\\S*,\\S*,\\S*,\\S*,\\S*,\\S*,(\\S+),\\S*,(\\S+),(\\S+),(\\d +);(\\d+),\\S*</regex>  
    <order>logsource,id,section,protocol,srcip,dstip,srcport, dstport</order>  
</decoder>
```

6.2 Custom Rules

A **rule** assigns severity levels and alerts based on matched events.

Example (pfsense-custom-rules.xml):

```

<group name="pfsense,firewall,">

  <!-- Generic pfSense log -->
  <rule id="100111" level="3">
    <decoded_as>pfsense-custom</decoded_as>
    <description>pfSense log received</description>
  </rule>

  <!-- Successful login -->
  <rule id="100112" level="3">
    <if_sid>100111</if_sid>
    <match>Successful login</match>
    <description>pfSense: Successful login from $(srcip)</ description>
  </rule>

  <!-- Allowed traffic -->
  <rule id="100113" level="5">
    <if_sid>100111</if_sid>
    <match>pass</match>
    <description>pfSense: Allowed traffic from $(srcip) to $( dstip)</description>
  </rule>

  <!-- Blocked traffic -->
  <rule id="100114" level="7">
    <if_sid>100111</if_sid>
    <match>block</match>
    <description>pfSense: Blocked traffic from $(srcip) to $( dstip)</description>
  </rule>

  <!-- Authentication error -->
  <rule id="100115" level="10">
    <if_sid>100111</if_sid>
    <match>authentication error</match>
    <description>pfSense: Authentication error for user $( dstuser) from
    $(srcip)</description> </rule>

</group>

```

- **Level 3 (Informational):** Successful logins.
- **Level 5 (Normal alert):** Allowed traffic.
- **Level 7 (Warning):** Blocked traffic.

- **Level 10 (High):** Authentication errors.

Conclusion

By combining **pfSense** and **Wazuh**, you now have a **practical SOC lab** that:

- Monitors and analyzes firewall traffic (allowed vs. blocked).
- Centralizes logs for correlation and alerting.
- Detects authentication attempts and suspicious behavior.
- Provides a foundation for hands-on practice in SOC operations.

This setup is ideal for **cybersecurity students, SOC analysts in training, or homelab enthusiasts** who want to explore **network security monitoring** in a safe environment.